

Mingqian Zhou

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RESEARCH INTERESTS

Embodied AI foundation models, vision-language-action models, world models, **reinforcement learning**, humanoid whole-body control.

EDUCATION

University of Cambridge

Cambridge, United Kingdom

Master of Philosophy in Advanced Computer Science

Sep. 2025 – Jun. 2026

- Relevant Coursework: Machine Visual Perception, Geometric Deep Learning, Computing for Collective Intelligence, Practical Research in Human-Centered AI

Northwestern Polytechnical University (NWPU)

Xi'an, China

Bachelor of Engineering in Robotics

Sep. 2021 – Jun. 2025

- Average score: 91.36/100 (**Rank: 1/345**)
- **Scholarships:** National Scholarship Nominee (2024); Outstanding Graduate of NWPU (2025); “Suzhou Yucui” Special Scholarship(2023); First-Class Scholarship (2022, 2023, 2024); “Kewei” Special Scholarship (2022)

Khalifa University (Exchange)

Abu Dhabi, UAE

Major in Electrical Engineering

Aug. 2023 – Dec. 2023

- **GPA:** 4.0/4.0 | Exchange selection ratio: 8/50

SELECTED RESEARCH & MANUSCRIPTS

GEAR: Gated Evidence-Aware Routing for Functional 3D Scene Graph Grounding

Mingqian Zhou, Dennis Rotondi, Andrea Giuseppe Di Francesco, Renjing Xu, and Pietro Lió. *Manuscript in preparation (target: IEEE RA-L)*.

Morphology-Consistent Humanoid Interaction through Robot-Centric Video Synthesis

W. Xu, ..., M. Zhou, ..., R. Xu. *Under review, CoRL 2026*.

RESEARCH EXPERIENCE

GEAR: Functional 3D Scene Graph Grounding

Cambridge, UK

Graduate Researcher (Supervisor: Prof. Pietro Lió)

Ongoing

- Independently designed GEAR’s full pipeline – candidate-conditioned evidence packets, structured graph prior, and a validation-frozen router for selective LLM verification.
- Built FunGraph-Bench, a 1,412-query benchmark from SceneFun3D and 3DSSG that tests whether a model can select the correct object from similar candidates using explicit scene-graph evidence.
- Matched graph-conditioned LLM cascades while invoking the verifier on only 58.1% of queries (0.7075 overall R@1).

Learning Cooperative Primitives for Multi-Task Multi-Agent Reinforcement Learning

Cambridge, UK

Graduate Researcher (Supervisor: Prof. Amanda Prorok)

Ongoing

- Developing a foundation-model-inspired **generalist policy** that learns reusable cooperative primitives for zero-shot transfer across unseen **multi-agent RL** tasks.
- Designing graph-based attentional communication modules for scalable decentralized execution and training multi-task curricula in VMAS.
- Benchmarking **RL** transfer and robustness in BenchMARL against MARL baselines such as MAPPO.

Morphology-Consistent Humanoid Interaction through Robot-Centric Video Synthesis

Guangzhou, China

Research Assistant (Prof. Renjing Xu)

Jan.–May 2026

- Led real-world deployment and validation for Dream2Act (CoRL 2026, under review) on a full-sized Unitree G1, executing zero-shot whole-body tasks from monocular input with 70% sitting and 60% kicking success.

- Owned the hardware experiment protocol, including scene setup, safety checks, reset procedures, trial logging, and repeatable success-rate measurement on physical runs.
- Contributed to sim-to-real evaluation by testing robot-centric hallucinated trajectories against human-centric retargeting baselines and documenting spatial-contact failures.

Animating Dynamic 3D Scenes from a Single Image

Cambridge, UK

Graduate Researcher (Supervisor: Prof. Cengiz Öztireli)

Oct. 2025 – Jan. 2026

- Explored single-image lifting into dynamic 4D scene representations, comparing implicit NeRF-style backbones with explicit 3D Gaussian Splatting for view synthesis quality.
- Integrated video diffusion priors to distill text-guided temporal dynamics into static 3D assets while preserving geometric consistency.
- Designed a CLIP-based evaluation and ablation pipeline for semantic alignment, motion realism, and texture fidelity under common generative artifacts.

Learning Human-to-Humanoid Real-Time Whole-Body Control

Guangzhou, China

Research Assistant (Supervisor: Prof. Renjing Xu – HKUST(GZ))

Jan. 2025 – Sep. 2025

- Built a **learning-based** whole-body control framework for real-time human-to-humanoid motion imitation, supporting high-precision teleoperation and VLM-driven autonomy.
- Constructed a scalable sim-to-data pipeline with a privileged motion imitator, retargeting large-scale human motion into physically feasible humanoid training data.
- Developed and evaluated a **Reinforcement Learning** policy for real-time motion imitation, achieving zero-shot **Sim-to-Real** transfer on a Unitree G1 for dynamic maneuvers (e.g., boxing, kicking).

Dynamic Reactive Motion Planning of Bipedal Robots

Xi'an, China

Research Assistant (Supervisor: Prof. Zhaohui Yuan)

Apr. 2023 – Jun. 2024

- Developed a hierarchical MPC/WBC framework for stable bipedal locomotion under external disturbances and unknown-environment constraints.
- Proposed a Linear Reaction Wheel Pendulum MPC for online gait generation and velocity tracking.
- Validated whole-body dynamic simulations and NAO hardware experiments using centroidal-Jacobian inverse kinematics and inverse dynamics.

AWARDS & HONOURS

National First Prize , Rescue Robot (Obstacle & Search-and-Rescue), 2022 China Robot Competition (RoboCup China)	Nov. 2022
National Second Prize , Rescue Robot (Autonomous Mapping), 2022 China Robot Competition (RoboCup China)	Nov. 2022
Outstanding Winner , Elite league of the National 3D competition	Aug. 2023
Meritorious Winner , Mathematical Contest in Modeling (MCM/ICM)	Apr. 2023
Star of Erudition	2023
Merit Student, NWPU	2022, 2023, 2024

EXTRACURRICULAR ACTIVITIES

Student Lead , Office of International Cooperation, NWPU	2021 – 2024
Coordinated student teams for multiple university international exchange programs; supported program organization, reception of visiting delegations, and on-campus liaison.	
Vice President , Cambridge Industrialization Society	Feb. 2026 – Now
Connected scholars and practitioners through conferences, seminars, and industry events; founded the 1st Cambridge Engineering Summit.	

SKILLS

Languages: English: Fluent (IELTS: 8.0), Mandarin: Native

Programming: Python, C++, MATLAB, $\text{\LaTeX}$

Learning & RL: PyTorch, reinforcement learning, multi-agent RL (MAPPO, BenchMARL, VMAS), sim-to-real transfer

Robotics: Isaac Gym, ROS, whole-body control, MPC/WBC, inverse kinematics & dynamics, teleoperation

Hardware: Unitree G1